

Cohort 5 Capstone

Prior Authorization Intake Workflow Agent

GitHub Repository

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Live Application

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Demo Video

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1. Problem Brief

Target User	Clinical reviewers and intake staff who process prior-authorization (PA) requests at health plans.
Pain Point	PA intake is manual, slow, and error-prone. Reviewers read each packet, spot missing fields, judge urgency, and route — all without consistent tooling. Mistakes delay patient care.
Value Hypothesis	An AI-assisted intake agent can flag incomplete packets, summarize clinical context, validate urgency tiers, and route cases — cutting review time and reducing routing errors.
Success Metric	Correct routing on all urgency tiers. 100% missing-field detection on synthetic cases. Zero AI-generated authorization verdicts — human gate always present.
Constraints	Synthetic data only (no PHI). Single API key, no external databases. Human approval mandatory before any decision is recorded.
Assumptions	Reviewers have browser access. Intake packets arrive as structured JSON (mimicking EHR export). Agent is decision-support — not a decision-maker.

2. Working Artifact

Stack	Python (FastAPI) + vanilla HTML/CSS/JS, SSE streaming, Claude Sonnet 4.6 via Anthropic API
Architecture	6-step pipeline — 3 AI steps (clinical judgment) + 3 rule-based steps (deterministic). All steps stream live progress to the browser.
Test Data	10 fully synthetic PA cases in data/sample_cases.json covering: clean baseline, missing fields, urgency tiers, triage mismatches, injection attacks, and bulk-missing edge cases.
Setup	Clone repo, set ANTHROPIC_API_KEY in .env, run <i>python main.py</i> , open localhost:8000. Under 5 minutes.
Deployment	Live on Render — auto-redeploys on every push. Free tier sleeps after 15 min of inactivity.

3. AI Evidence

Model and Tool Choices

Model: Claude Sonnet 4.6 via Anthropic API. Three of six workflow steps use AI — chosen because they require clinical judgment that cannot be reliably encoded as rules. The remaining three steps are deterministic (same input, same output every time).

Step	Type	Rationale
1. Field completeness check	Rule-based	Binary check — field is present or missing. No judgment needed.
2. Clinical summarization	Claude Sonnet 4.6	Synthesizes free-text clinical notes into a concise summary.
3. Follow-up questions	Claude Sonnet 4.6	Identifies missing information and phrases follow-up questions.
4. Urgency validation	Claude Sonnet 4.6	Cross-checks declared urgency against clinical narrative.
5. Case routing	Rule-based	Deterministic routing table — same input always gives same output.
6. Recommendation memo	Rule-based	Template assembled from steps 1–5 outputs. No AI needed.

Evaluation Suite

72 offline assertions run in under 1 second with no API key:

- Field completeness — all required fields across all 10 synthetic cases
- Input safety scanner — injected instructions caught and refused
- Routing table — STAT / Urgent / Routine all produce correct SLAs
- Urgency mismatch detection — both under-triage and over-triage caught
- Memo determinism — same inputs always produce the same memo text

Live eval suite calls the model and reports observed rates across 3 samples per case. Containment properties are hard-failed: the agent must never obey injected instructions and never emit an authorization verdict.

Known Failure Modes and Risks

- **Summarization nuance:** May miss nuance in very short notes — mitigated by mandatory human review on all AI steps.
- **Urgency disagreement:** AI may disagree with declared tier — reviewer sees the rationale and can override.
- **Non-determinism:** Model responses vary across runs — live eval samples 3 runs per case to measure variance.
- **Injection evasion:** Scanner uses keyword + heuristic matching — a sophisticated injection might evade it; case is forced to human review anyway.

Human-in-the-Loop Controls

- Every case ends at a **human approval gate** — no authorization decision is recorded without explicit reviewer action.
- Reviewers can **override any AI decision**; all previous decisions preserved in audit trail with name, role, and timestamp.
- Cases flagged by input-safety scanner are **automatically routed to human review** — the pipeline halts.

4. Enterprise Readiness

Area	Current Build	Production Path
Data Classification	All data is fully synthetic — no PHI, no production data. sample_cases.json contains no real patient information.	Intake packets classified as PHI/PII, subject to HIPAA controls.
Access & Auth	Role-based login with 3 reviewer accounts (Clinical Review Lead, Senior Clinical Reviewer, Clinical Ops Manager). Every action tagged with identity.	SSO / LDAP integration with role-based permissions.
Audit & Logging	Full audit trail: every pipeline run, decision, and override stored with actor identity and timestamp. PDF export includes all metadata.	Logs forwarded to SIEM; audit table persisted to compliant database.
Security	Input safety scanner strips markup and length-bounds text before AI calls. API key via env variable (.env gitignored).	Azure Key Vault for secrets, network isolation, TLS termination.
Handoff	Built by Jayanth Dolai. Codebase documented, dependencies pinned in requirements.txt, deployable via single Render service.	New engineer can clone, set API key, and run in under 5 minutes.